



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

UNIT TEST

18-07-2021

JEE(Main + Advanced) : Enthusiast Course (PHASE : TEAS)

Test Type : UNIT TEST

ANSWER KEY

PHYSICS

SEC-1	Q.	1	2	3	4	5	6
	A.	A,C	A,D	B	A,C	B	A,D
SEC-2	Q.	1	2	3	4	5	6
	A.	4.00	5.00	1.00	0.20	1.11	1.00
SEC-3	Q.	1	2	3	4	5	6
	A.	3	1	2	8	2	3

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6
	A.	B,C,D	A,B,D	B,C,D	A,B,C,D	B,D	A,B,C
SEC-2	Q.	1	2	3	4	5	6
	A.	5.00	3.00	6.00	2.53	15.00	375.00
SEC-3	Q.	1	2	3	4	5	6
	A.	7	4	2	0	4	4

MATHEMATICS

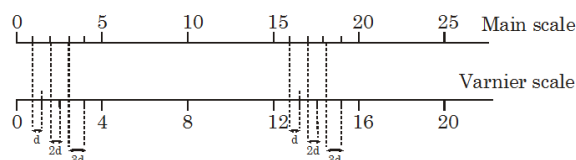
SEC-1	Q.	1	2	3	4	5	6
	A.	A,C,D	A,B,C	A,C	A,B	A,C,D	A,D
SEC-2	Q.	1	2	3	4	5	6
	A.	9.86	6.40	0.00	100.00	1.57	16.00
SEC-3	Q.	1	2	3	4	5	6
	A.	7	0	0	0	0	3

HINT – SHEET

PHYSICS

SEC-1

1. Ans (A,C)



$$d = 0.25 \text{ mm}$$

$$LC = 0.25 \text{ mm}$$

$$\text{zero error} = -0.75 \text{ mm}$$

while taking the reading zero of vernier will be between 9th and 10th of main scale division.

$$\text{Reading} = 9 \text{ mm} + 0.50 \text{ mm} + 0.75 \text{ mm} = 10.25 \text{ mm}$$

2. Ans (A,D)

$$U = \frac{1}{2} kx^2$$

$$= kx \times \frac{x}{2}$$

$$= P_0 A \times \frac{V_0}{A} = P_0 V_0$$

$$V = A \times \frac{x}{2}$$

$$P = \frac{kx}{A}$$

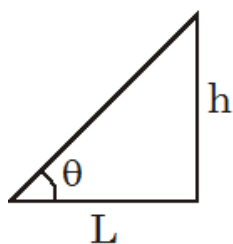
$$\frac{P}{V} = \text{constant}$$

$$C = \frac{5R}{2} + \frac{R}{2} = 3R$$

$$Q = nC\Delta T = 3nR \times 2T_0$$

$$Q_T = 2Q = 6P_0 V_0$$

3. Ans (B)



$$\frac{h}{L} = \tan \theta$$

$$\ln h = \ln L + \ln \tan \theta$$

$$\begin{aligned} \frac{dh}{h} &= \frac{dL}{L} + \frac{d\theta}{\sin \theta \cos \theta} \\ &= \frac{dL}{L} + \frac{2d\theta}{\sin^2 \theta} \\ &= \frac{0.5}{100} + \frac{4x}{\sqrt{3}} \frac{0.2\pi}{1.80} = 1.3\% \end{aligned}$$

4. Ans (A,C)

$$\begin{aligned} \frac{1}{v} - \frac{1}{u} &= \frac{1}{8} \\ \frac{1}{u} - \frac{1}{-25} &= \frac{1}{50} \\ \frac{1}{u} &= \frac{1}{50} - \frac{1}{25} = -\frac{1}{50} \end{aligned}$$

$$D = 101 + 50 = 100 \text{ cm}$$

$$\frac{h_I}{h_0} = \frac{v}{u} = \frac{-50}{-25} = 2$$

$$h = 0.5 \text{ mm} \Rightarrow h_I = 1 \text{ mm}$$

$$\frac{d}{2} = h_I - 0.5 = 0.5 \text{ mm}$$

$$\omega = \frac{\lambda D}{d} = 0.6 \text{ mm}$$

5. Ans (B)

$$N = N_0 e^{-\lambda t}$$

$$\frac{dN}{dt} = -N_0 \lambda e^{-\lambda t}$$

$$\Rightarrow N_0 \lambda = \frac{N_0}{T}$$

$$\Rightarrow \frac{1}{t_{av}} = \frac{1}{T}$$

$$T_{1/2} = \frac{\ln 2}{\lambda} = T \ln 2$$

PHYSICS

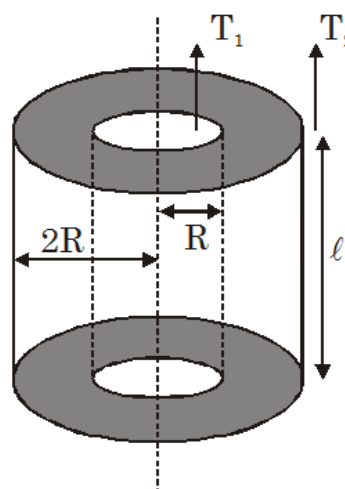
SEC-2

1. Ans (4.00)

$$H = -K \cdot 2\pi r \ell \frac{dT}{dr}$$

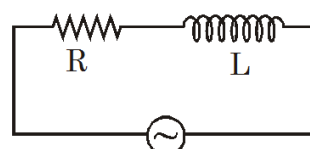
$$\int_{R_1}^{R_2} \frac{H dr}{2\pi r \ell} = -K \int_{T_1}^{T_2} dT$$

$$\Rightarrow H = \frac{2\pi \ell k (T_1 - T_2)}{\ln \frac{R_2}{R_1}} H_1 = H_f$$



2. Ans (5.00)

Coil has internal resistance r.



$$40^2 = (ix_L)^2 + (ir)^2$$

$$50^2 = (ix_L)^2 + (i(r + 50))^2$$

$$i = \frac{20}{50} = 0.4$$

$$900 = 0.16[100r + 2500]$$

$$900 = 16\pi + 400$$

$$r = \frac{500}{16} \Omega$$

$$P = i^2 r$$

$$= 0.16 \times \frac{500}{16} = 5 \text{ W}$$

3. **Ans (1.00)**

For maxima at P, path difference = $n\lambda$, n = integer

$$\Rightarrow (s_2P - s_1P) - (\mu - 1)t = n\lambda$$

$$\Rightarrow y \frac{d}{D} - (\mu - 1)t = n\lambda$$

$$\Rightarrow n\lambda = 10^{-6} \text{ m} \Rightarrow n = \frac{10^{-6}}{\lambda}$$

$$n^{\max} = \frac{10^{-6}}{4000 \times 10^{-10}} = 2.5 \text{ \&}$$

$$n^{\min} = \frac{10^{-6}}{7000 \times 10^{-10}} = 1.42$$

$\therefore n = 2$, $\therefore$ only one wavelength will give maxima at P for $n = 2$.

4. **Ans (0.20)**

I_L and I_C will be in opposite phases.

$$I_{\text{net}} = I_L - I_C$$

$$= 0.6 - 0.4 = 0.2 \text{ A}$$

5. **Ans (1.11)**

$$Q = \Delta m C^2 = \frac{hC}{\lambda}$$

$$Mv = \frac{h}{\lambda} = \frac{Q}{C}$$

$$v = \frac{Q}{MC} = \frac{0.0003 \times C^2}{81 \times C}$$

$$\frac{3 \times 3 \times 10^{+5}}{8 + 9} = 1.11 \text{ km/s}$$

6. **Ans (1.00)**

$$R = \frac{mv}{qB} = \frac{mE}{qB^2}$$

$$R_{\text{aw}} = 0.100 \text{ mm}, \Delta R = 0.005 \text{ mm}$$

$$S = \frac{q}{m} = \frac{E}{RB^2} = \frac{100}{10^{-4} \times 0.01} = 10^8 \text{ C/kg}$$

$$\frac{\Delta S}{S} = \frac{\Delta E}{E} + \frac{\Delta R}{R} + \frac{2\Delta B}{B}$$

$$\Delta S = 10^8 \left[\frac{1}{100} + \frac{2}{100} \times 2 + \frac{0.005}{0.100} \right] = 10^7$$

PHYSICS

SEC-3

1. **Ans (3)**

$$dQ = -2dU$$

$$nC_dT = -2nC_vdT$$

$$C = -2C_v = -\frac{2R}{r-1}$$

$$C = C_v - \frac{R}{m-1}$$

$$\frac{R}{m-1} = \frac{3R}{r-1}$$

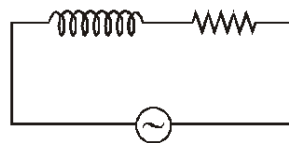
$$m = \frac{r-1}{3} + 1 = \frac{r+2}{3}$$

$$TV^{m-1} = C$$

$$m-1 = \frac{r+2}{3} - 1 = \frac{r-1}{3}$$

$$k = 3$$

2. **Ans (1)**



P same $\Rightarrow Z$ = same

$$\Rightarrow \omega L = \frac{1}{\omega C} - \omega L$$

$$2\omega L = \frac{1}{\omega C}$$

$$C = \frac{1}{2\omega^2 L} = \frac{1}{2r4\pi^2 \times 2500 \times 5}$$

$$= \frac{10^{-5}}{2} = 5 \times 10^{-6} \text{ F}$$

3. **Ans (2)**

$$i_1 = \frac{40}{20} \left(1 - e^{-\frac{20t}{10^{-2}}} \right) = 1.5 \text{ A}$$

$$i_2 = \frac{40}{40} e^{-\frac{t}{40 \times 25 \times 10^{-6}}} = 0.5 \text{ A}$$

$$i_T = 2 \text{ A}$$

4. **Ans (8)**

$$P = \frac{E^2}{R} = \frac{\left(B\omega \frac{\ell}{2} \right)^2}{R}$$

$$= \frac{\left(\frac{1 \times 2 \times 4}{2} \right)^2}{1} = 16 \text{ W}$$

5. Ans (2)

$$K = \frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi$$

$$\frac{1}{2} \times 9 \times 10^{-31} v^2 = 4 - 2.2 = 1.8 \times 1.6 \times 10^{-19}$$

$$v = \sqrt{\frac{4.8 \times 2}{9}} \times 1.6 \times 10^{10} = 8 \times 10^5 \text{ m/s}$$

$$v_1^2 = v^2 - 2ad = v^2 - qV$$

$$v_4 k_f = 1.8 - 1 = 0.8 \text{ eV}$$

$$v_f = \sqrt{\frac{2 \times 0.8 \times 1.6 \times 10^{-19}}{9 \times 10^{-31}}}$$

$$= \frac{1.6}{3} \times 10^6 = \frac{16}{3} \times 10^5 \text{ m/s}$$

$$S = \left(\frac{v+u}{2} \right) t$$

$$S = \frac{10}{2} \left(\frac{16}{3} + 8 \right) \times t$$

$$= 10^5 \times \frac{20}{3} \times 3 \times 10^{-9}$$

$$2 \times 10^{-3} \text{ m} = 2 \text{ mm}$$

6. Ans (3)

$$\ln x = \ln(a^2 - b^2) = \frac{1}{2} \ln(c + d)$$

$$\frac{dx}{x} = \frac{2adx + 2bdb}{a^2 - b^2} + \frac{1}{2} \frac{dc + dd}{c + d}$$

$$= \frac{2 \times 3 \times 10^{-2} + 2 \times 5 \times 10^{-2}}{16} + \frac{1}{2} \times \frac{0.64}{16}$$

$$= (10^{-2} + 0.02) \times 100 = 3\%$$

CHEMISTRY

SEC-1

2. Ans (A,B,D)

CHEMISTRY

SEC-2

5. Ans (15.00)

$$\text{Ionization constant} : 8 \times 10^{-7}$$

$$a = 8, b = 7, a + b = 15$$

CHEMISTRY

SEC-3

5. Ans (4)

$$E = E^\circ - 0.06 \text{ pH}$$

$$0.22 = 0.46 - 0.06 \text{ pH}$$

$$0.06 \text{ pH} = 0.46 - 0.22$$

MATHEMATICS

SEC-1

1. Ans (A,C,D)

$$\text{for A,B } \int x \left(1 - \frac{1}{x^4} \right)^{1/5} \frac{1}{x^6} dx$$

$$\text{let } 1 - \frac{1}{x^4} = t \Rightarrow dt = \frac{4}{x^5} dx$$

$$\int t^{1/5} \frac{dt}{4} = \frac{1}{4} \frac{t^{6/5}}{6/5} = \frac{5}{24} t^{6/5}$$

$$(C) \int (\tan^2 x + 1) (\tan x + 1) dx$$

$$\frac{(\tan x + 1)^2}{2} + C$$

$$(D) \int (\tan x \sec^2 x + \sec^2 x) dx$$

$$\frac{1}{2} \tan^2 x + \tan x + C$$

$$\text{or } \frac{\sec^2 x}{2} + \tan x + \left(C - \frac{1}{2} \right)$$

2. Ans (A,B,C)

For nil potent matrix of order 2, $A^2 = 0$

$$A(I + A) = A + A^2 = A + 0 = A$$

for $n \geq 2$

$$A(I + A)^n = A[I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n]$$

$$\Rightarrow A(I + A) = A + A^2[{}^nC_1 I + {}^nC_2 A + \dots] = A$$

similarly for $A(I - A)^n$

For idempotent matrix,

$$A^2 = A = A^3 = \dots = A^n$$

$$(I + A)^n = I + {}^nC_1 A + {}^nC_2 A^2 + \dots + {}^nC_n A^n$$

$$= I + A(2^n - 1)$$

3. Ans (A,C)

$$\text{Slope of tangent} = -\frac{y}{2}$$

$$\frac{dy}{dx} = -\frac{y}{2} \Rightarrow y = 2e^{-\frac{x}{2}}$$

4. Ans (A,B)

The differential equation of

$$F : \frac{dy}{dx} + y = x + 1$$

The differential equation of the orthogonal trajectories of F is

$$\frac{-dx}{dy} + y = x + 1$$

$$\text{i.e., } \frac{dx}{dy} + x = y - 1, \text{ which is linear in } x.$$

its general solution is

$$xe^{\int 1 dy} = \int e^{\int 1 dy} (y - 1) dy + c$$

$$\text{i.e., } xe^y = \int e^y (y - 1) dy + c$$

$$\text{i.e., } xe^y = e^y(y - 2) + c$$

The particular solution is $x = y - 2$

5. Ans (A,C,D)

$$A_n = \sum_{t=1}^n \frac{n}{n^2 + nt + t^2} \text{ put } k + 1 = t$$

$$B_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kn + k^2},$$

$$\text{now on solve } A_n < \frac{\pi}{3\sqrt{3}} < B_n$$

6. Ans (A,D)

$$AB = A \Rightarrow ABA = A^2 \Rightarrow A^2 = A$$

$$\text{Similarly } B^2 = B$$

A and B are idempotent

$$\text{Also } (A - B)^2 = (A - B)(A - B)$$

$$= A^2 - AB - BA + B^2 = A - A - B + B = 0$$

$$\therefore (A - B) \text{ is nilpotent.}$$

Correct options are (A) and (D).

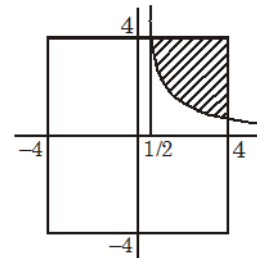
MATHEMATICS

SEC-2

1. Ans (9.86)

$$|x| \leq 4 \text{ and } |y| \leq 4$$

Area of shaded portion

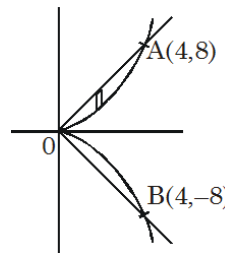


$$\Delta = \int_{1/2}^4 \frac{2}{x} dx \Rightarrow [2 \ln x]_{1/2}^4$$

$$\Rightarrow 2 [\ln 4 + \ln 2] = \ln 64$$

$$\text{Shaded region} = 14 - \ln 64$$

2. Ans (6.40)



Required area

$$= 2 \int_0^4 (2x - x^{3/2}) dx = \frac{32}{5}$$

3. Ans (0.00)

$$\text{Using } \int_0^1 f(x) dx = \int_0^1 f(1-x) dx$$

$$I = \int_0^1 \sqrt[3]{x^2(2x-3) + (1-x)} dx$$

$$= \int_0^1 \sqrt[3]{(1-x)^2(-1-2x) + x} dx$$

$$= - \int_0^1 \sqrt[3]{(x^2 - 2x + 1)(1 + 2x) - x} dx$$

$$= - \int_0^1 \sqrt{2x^3 - 3x^2 - x + 1} dx = -I$$

$$2I = 0 \quad \therefore I = 0$$

MATHEMATICS

SEC-3

4. Ans (100.00)

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow A^n = \begin{cases} I & n \text{ is even} \\ A & n \text{ is odd} \end{cases}$$

$$\sum_{r=1}^{100} A^r = (A + A^3 + A^5 + \dots + A^{99}) + (A^2 + A^4 + \dots + A^{100})$$

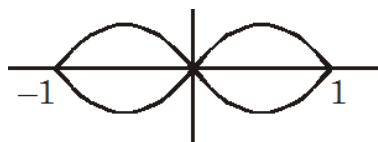
5. Ans (1.57)

$$y^2 = -x^6 + x^2$$

$$y = \pm x \sqrt{-x^4 + 1}$$

$$y = x \sqrt{-x^4 + 1}$$

$$y = -x \sqrt{-x^4 + 1}$$



$$x^2 = t$$

$$= 2 \int_0^1 \sqrt{1-t^2} dx$$

$$= 2 \left(\frac{t}{2} \sqrt{1-t^2} + \frac{1}{2} \sin^{-1} t \right)_0^1$$

$$= \left(0 + \frac{\pi}{2} \right) - (0 - 0) = \frac{\pi}{2}$$

6. Ans (16.00)

$$y = 3 + \frac{mx + n - 3}{1 + x^2}$$

now $mx + n - 3 < 0$ Think !

$$m = 0 \quad \forall x \in \mathbb{R}$$

$$n - 3 < 0$$

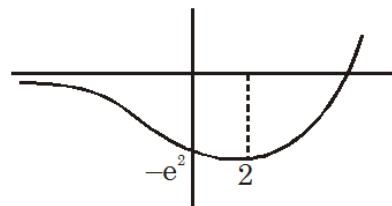
$$y = 3 + \frac{n - 3}{1 + x^2}$$

$y_{\min}$ at $x = 0$ think !

$$y = 3 + n - 3 = -4$$

$$n = -4$$

1. Ans (7)



$$e^x(x-3) = k$$

$$\text{Let } f(x) = e^x(x-3)$$

$$f'(x) = e^x(x-2)$$

$$k \in (-e^2, 0)$$

2. Ans (0)

Function satisfying given rule :

$$f(x) = \frac{x^2}{4}$$

Least area occurs for $a = 0$.

3. Ans (0)

$$g(x) > 0 \Rightarrow f(x) > 0$$

$$g\left(\frac{1}{2}\right) = \frac{1}{e} \Rightarrow f\left(\frac{1}{2}\right) = 1$$

$$\frac{dg(x)}{dx} < 0 \Rightarrow f'(x) < 2f(x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} < 2$$

$$\Rightarrow \int_{1/2}^x \frac{f'(x)}{f(x)} dx < \int_{1/2}^x 2 dx$$

$$\therefore 0 < f(x) < e^{2x-1}$$

$$\int_{1/2}^1 0 dx < \int_{1/2}^1 f(x) dx < \int_{1/2}^1 e^{2x-1} dx$$

$$0 < \int_{1/2}^1 f(x) dx < \frac{e-1}{2}$$

4. Ans (0)

Given $1 \leq f(x) + f'(x) \leq 2 \forall x \in \mathbb{R}$

Mat. by e^x and integrating between limits 0 to 1

$$\int_0^1 e^x dx \leq \int_0^1 e^x (f(x) + f'(x)) dx \leq \int_0^1 2e^x dx$$

$$(e - 1) \leq (e^x f(x))_0^1 \leq 2(e - 1)$$

$$1 - \frac{1}{e} \leq f(1) \leq 2 - \frac{2}{e}$$

5. Ans (0)

$$I = \int \frac{x^2 + 20}{(x \sin x + 5 \cos x)^2} dx \text{ multiply \& div}$$

by x^8

$$I = \int \frac{x^5 \cdot x^3 (x^2 + 20)}{(x^5 \sin x + 5x^4 \cos x)^2} dx$$

$$I = \int \frac{\left(\frac{x^5}{\cos x}\right)}{I} \frac{x^3 (x^2 + 20) \cos x}{\frac{(x^5 \sin x + 5x^4 \cos x)^2}{II}} dx$$

$$x^5 \sin x + 5x^4 \cos x = t$$

$$[x^5 \cos x + \sin x 5x^4 + 5x^4 (-\sin x) + 5 \cos x$$

$$x \cdot 4x^3] dx = dt$$

$$x^3 \cos x (x^2 + 20) dx = dt$$

$$I = -\frac{x^5}{\cos x} \frac{1}{(x^5 \sin x + 5x^4 \cos x)} + \int \frac{\cos x \cdot 5x^4 + x^5 \sin x}{(x^5 \sin x + 5x^4 \cos x)} \sec^2 x dx$$

$$I = -\frac{x}{\cos x (x \sin x + 5 \cos x)} + \tan x + c$$

6. Ans (3)

$$\int \frac{(3x^2 + 2x) dx}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5}$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x^3 + x^2 + 1}{2} \right) + C$$

$$\therefore a = 2 \text{ and } g(1) = \frac{3}{2}, g(0) = \frac{1}{2}$$